

Area under a polar curve

Basic skills

Q1. a) $r=1$ so

$$\frac{1}{2} \int_0^{2\pi} r^2 d\theta = \frac{1}{2} \int_0^{2\pi} 1 d\theta = \frac{1}{2} [\theta]_0^{2\pi} = \pi$$

If $r = \text{const}$ can use πr^2 as $\frac{1}{2} [\text{const}^2 \theta]_0^{2\pi} = \pi \text{const}^2$

b) $r=2$ $\pi \times 2^2 = 4\pi$

c) $r=5$ $\pi \times 5^2 = 25\pi$

d) $r=10$ $\pi \times 10^2 = 100\pi$

Q2. $r = \sqrt{\theta}$, $0 \leq \theta < 2\pi$

$$\begin{aligned} \frac{1}{2} \int_0^{2\pi} (\sqrt{\theta})^2 d\theta &= \frac{1}{2} \int_0^{2\pi} \theta d\theta = \frac{1}{2} \left[\frac{1}{2} \theta^2 \right]_0^{2\pi} \\ &= \frac{1}{2} \left(\frac{1}{2} (2\pi)^2 - 0 \right) = \frac{1}{4} \times 4\pi^2 = \pi^2 \end{aligned}$$

Q3. a) $r = \theta$, $0 \leq \theta \leq \pi$

$$\begin{aligned} \frac{1}{2} \int_0^{2\pi} \theta^2 d\theta &= \frac{1}{2} \left[\frac{1}{3} \theta^3 \right]_0^{2\pi} = \frac{1}{6} ((2\pi)^3 - 0) \\ &= \frac{8}{6} \pi^3 \end{aligned}$$

b) $r = \theta^2$, $0 \leq \theta \leq \pi$

$$\frac{1}{2} \int_0^{2\pi} (\theta^2)^2 d\theta = \frac{1}{2} \int_0^{2\pi} \theta^4 d\theta = \frac{1}{2} \left[\frac{1}{5} \theta^5 \right]_0^{2\pi}$$

$$= \frac{1}{10} \left((2\pi)^5 - 0 \right) = \frac{32\pi^5}{10}$$

Competency skills

Q4. $r = R$ constant

$$\frac{1}{2} \int_0^{2\pi} R^2 d\theta = \frac{1}{2} \left[R^2 \theta \right]_0^{2\pi} = \frac{1}{2} R^2 \times 2\pi$$

$$= \pi R^2$$

Q5. a) $r = \sin \theta$, $0 \leq \theta \leq \pi$

$$\frac{1}{2} \int_0^{\pi} \sin^2 \theta d\theta = \frac{1}{2} \int_0^{\pi} \frac{1}{2} - \frac{1}{2} \cos 2\theta d\theta$$

$$\cos 2\theta = 1 - 2\sin^2 \theta$$

$$= \frac{1}{2} \left[\frac{1}{2} \theta - \frac{1}{4} \sin 2\theta \right]_0^{\pi} = \frac{1}{2} \left(\left(\frac{\pi}{2} - 0 \right) - (0 - 0) \right)$$

$$= \frac{1}{4} \pi$$

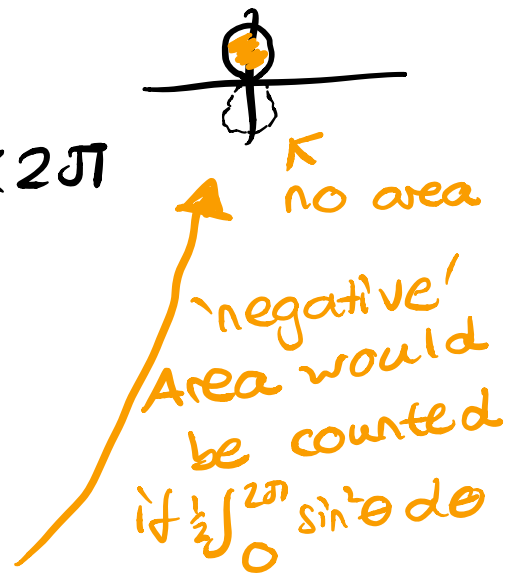
b) $r = \sin \theta$, $0 \leq \theta < 2\pi$

but $r < 0$ for $\pi < \theta < 2\pi$

so

$$\frac{1}{2} \int_0^\pi \sin^2 \theta \, d\theta = \frac{1}{4}\pi$$

→
see a)



c) Because $\sin \theta < 0$ for $\pi < \theta < 2\pi$

Q6. $r = 2 + \cos \theta$, $0 \leq \theta < 2\pi$

$$\frac{1}{2} \int_0^{2\pi} (2 + \cos \theta)^2 \, d\theta$$

never negative

$$= \frac{1}{2} \int_0^{2\pi} (4 + 4 \cos \theta + \cos^2 \theta) \, d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} (4 + 4 \cos \theta + \frac{1}{2} + \frac{1}{2} \cos 2\theta) \, d\theta$$

↙ $\cos 2\theta = 2\cos^2 \theta - 1$

$$= \frac{1}{2} \left[4\theta + 4 \sin \theta + \frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right]_0^{2\pi}$$

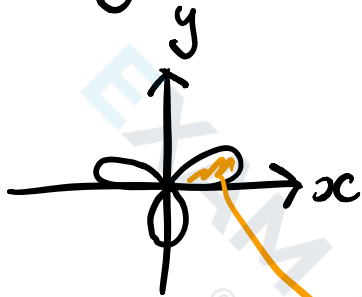
$$= \frac{1}{2} \left((8\pi + 0 + \pi + 0) - 0 \right) = \frac{9\pi}{2}$$

Q7. $r = \sin 3\theta$, $-\pi < \theta \leq \pi$

$r > 0$ for $0 \leq \theta \leq \frac{\pi}{3}$, $\frac{2\pi}{3} \leq \theta \leq \pi$,

$-\frac{2\pi}{3} \leq \theta \leq -\frac{\pi}{3}$

3 Regions, but can use symmetry as each region is the same



so $3 \times \frac{1}{2} \int_0^{\frac{\pi}{3}} \sin^2 3\theta d\theta$

Area of one region

one region:

$\frac{1}{2} \int_0^{\frac{\pi}{3}} \sin^2 3\theta d\theta$ 'double angle'

$\cos 6\theta = 1 - 2\sin^2 3\theta$

$= \frac{1}{2} \int_0^{\frac{\pi}{3}} \frac{1}{2} - \frac{1}{2} \cos 6\theta d\theta = \frac{1}{2} \left[\frac{1}{2} \theta - \frac{1}{12} \sin 6\theta \right]_0^{\frac{\pi}{3}}$

$= \frac{1}{2} \left(\left(\frac{\pi}{6} - 0 \right) - 0 \right) = \frac{\pi}{12}$

so $3 \times \frac{\pi}{12} = \frac{\pi}{4}$ is full area.

Advanced skills

Q8. $\theta = \ln(r)$, $1 \leq r \leq e^{2\pi}$ rearrange for $r(\theta)$

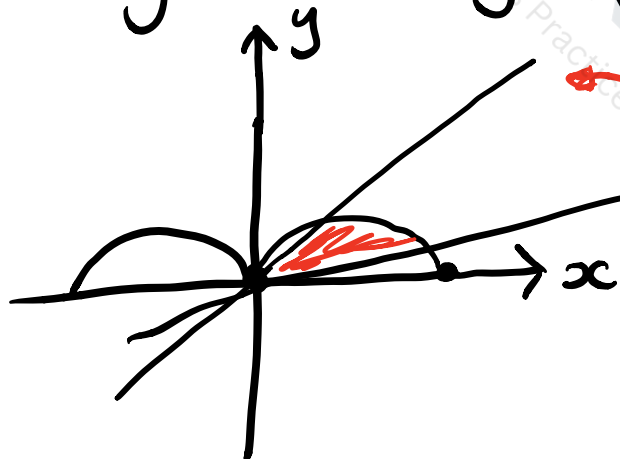
$$r = e^{\theta}, \quad 0 \leq \theta \leq 2\pi$$

$$\text{so } \frac{1}{2} \int_0^{2\pi} (e^{\theta})^2 d\theta = \frac{1}{2} \int_0^{2\pi} e^{2\theta} d\theta$$

$$= \frac{1}{2} \left[\frac{1}{2} e^{2\theta} \right]_0^{2\pi} = \frac{1}{2} \left(\frac{1}{2} e^{4\pi} - \frac{1}{2} \right)$$

$$= \frac{1}{4} (e^{4\pi} - 1)$$

Q9. $y = \sqrt{3}x$, $y = \frac{1}{\sqrt{3}}x$, $r = 2 - 2\sin\theta$
 $0 \leq \theta < \pi$



need to find θ
representations
of lines

$$\frac{y}{x} = \sqrt{3}, \quad \frac{y}{x} = \frac{1}{\sqrt{3}}$$

$$\theta = \tan^{-1}(\sqrt{3}) \quad \theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$\theta = \frac{\pi}{3}$$

$$\theta = \frac{\pi}{6}$$

use as limits

so $\frac{1}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{6}} (2-2\sin\theta)^2 d\theta$

$$= \frac{1}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{6}} (4 - 8\sin\theta + 4\sin^2\theta) d\theta$$

$\checkmark \cos 2\theta = 1 - 2\sin^2\theta$

$$= \frac{1}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{6}} (4 - 8\sin\theta + 2 - 2\cos 2\theta) d\theta$$

$$= \frac{1}{2} \left[4\theta + 8\cos\theta + 2\theta - \sin 2\theta \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$$

$$= \frac{1}{2} \left(\left(\frac{4\pi}{3} + 4 + \frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right) - \left(\frac{4\pi}{6} + 4\sqrt{3} + \frac{2\pi}{6} - \frac{\sqrt{3}}{2} \right) \right)$$

$$= \frac{1}{2} \left((2\pi + 4) - (\pi + 4\sqrt{3}) \right)$$

$$= \frac{1}{2} (\pi + 4 - 4\sqrt{3}) = \frac{1}{2}\pi + 2(1 - \sqrt{3})$$

Q10. $y = x$, $y = 0$ (+ve x), $y = x\sqrt{x^2 + y^2}$

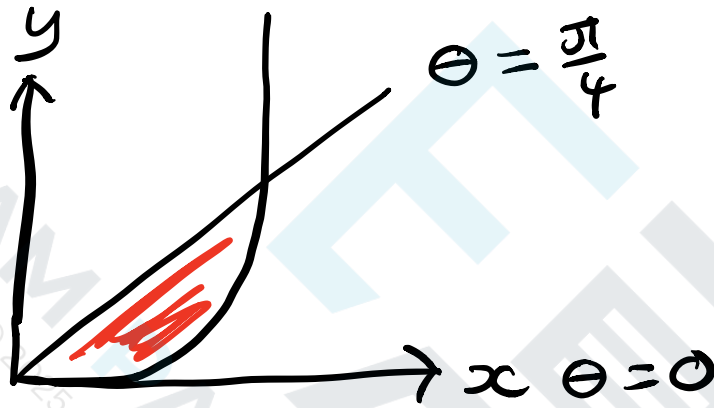
convert $\checkmark \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}(1) = \frac{\pi}{4}$

$\theta = \frac{\pi}{4}$ $\theta = 0$

for $y = x\sqrt{x^2 + y^2}$ $\frac{y}{x} = \sqrt{r^2}$

$$\frac{y}{x} = \frac{r \sin \theta}{r \cos \theta} = \tan \theta$$

So Polar curve is $r = \theta$



$$\text{so } \frac{1}{2} \int_0^{\pi/4} \tan^2 \theta \, d\theta = \frac{1}{2} \int_0^{\pi/4} \sec^2 \theta - 1 \, d\theta$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$= \frac{1}{2} \left[\tan \theta - \theta \right]_0^{\pi/4}$$

$$= \frac{1}{2} \left(\left(\tan \frac{\pi}{4} - \frac{\pi}{4} \right) - 0 \right) = \frac{1}{2} \left(1 - \frac{\pi}{4} \right)$$

Q11. Yes, consider $r = \theta$

for $0 \leq \theta < 2\pi$ & $-\pi < \theta \leq \pi$

None periodic functions
are the issue!

less area